

An Explicit Obstruction to Uniform Two-Word π -Representability

Lennart Rudolph 

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Abstract

Adamson, Dietz, Fleischmann, Huch, and Sacher introduced the hierarchy $\mathcal{G}_k$ of graphs represented by two k -uniform words under equality of two-letter projections. They proved that no fixed level contains every graph and asked for an explicit graph outside $\mathcal{G}_2$. We first recast the word condition: $G \in \mathcal{G}_k$ if and only if $\overline{G}$ is the image of a permutation graph under a graph compaction whose fibres all have size k . Equivalently, G is obtained from a dimension-at-most-two poset partitioned into k -element chains, with two distinct quotient vertices adjacent when the union of their chains is a chain. Using the two-order form of this characterization, we encode every outside-neighborhood trace on an m -vertex set by two weak compositions and obtain the bound $\binom{k(m+1)}{k}^2$. For $k = 2$ and $m = 20$ this bound is 741321, so the bipartite graph with parts $[20]$ and $\{0, \dots, 741321\}$, where i is adjacent to j exactly when bit i of j is 1, lies outside $\mathcal{G}_2$. The same trace estimates place the standard open-neighborhood VC-dimension of $\mathcal{G}_2$ between 10 and 19. We quantify the gap between this explicit construction, a 35-vertex counting argument, and exact small trace capacities. A reproducible artifact supplies independent exact checks and a scoped Lean 4 formalization of the order bridge and finite arithmetic.

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1 Introduction

For a word w and a set S of letters, let $\pi_S(w)$ be the word obtained by deleting every letter not in S . We abbreviate $\pi_{\{x,y\}}$ to $\pi_{x,y}$. A word is k -uniform if every letter in its alphabet occurs exactly k times. Given two words w, v with the same alphabet V , let $G(w, v)$ be the simple graph on V in which, for distinct x, y ,

$$xy \in E(G(w, v)) \iff \pi_{x,y}(w) = \pi_{x,y}(v).$$

Write $\mathcal{G}_k$ for the class of graphs $G(w, v)$ obtained from two k -uniform words.

Adamson, Dietz, Fleischmann, Huch, and Sacher introduced these classes and proved that $\mathcal{G}_1$ is the class of permutation graphs, that $\mathcal{G}_k$ is hereditary, and that $\mathcal{G}_k \subseteq \mathcal{G}_{k+1}$ for every k [1, 2]. They also proved $\bigcup_{k \geq 1} \mathcal{G}_k$ to be the class of all finite graphs, while no fixed $\mathcal{G}_k$ contains every graph [1, Thm. 29].¹ The v1 conclusion asks for a graph outside $\mathcal{G}_2$ [1, p. 19].

¹HTML uses section numbering; the SHA-pinned v1 PDF calls this Theorem 29.

The purpose of this note is twofold. First, we identify the uniform hierarchy with a prescribed-fibre compaction problem for permutation graphs and, in parallel, with a chain-contraction problem for dimension-two posets. This is a reformulation rather than a new graph class, but it exposes the local parameter needed for the second contribution. Namely, the positions of a fibre relative to a fixed collection of fibres in each of the two realizing orders give an explicit polynomial-size code for its neighborhood trace. A family of more traces than codes then gives a completely specified graph outside $\mathcal{G}_2$.

No prior resolution or substantially equivalent characterization was found in the documented search through 18 August 2026. This finite search does not establish an absolute priority claim and should be repeated if public posting is delayed.

2 A fixed-fibre compaction characterization

Let $[k] = \{1, \dots, k\}$. If w is a k -uniform word on V , label the i -th occurrence of x by the token (x, i) . Reading w gives a total order L_w on

$$T = V \times [k].$$

For two k -uniform words w, v , let $H(w, v)$ be the graph on T in which two tokens are adjacent exactly when their relative orders in L_w and L_v disagree. Thus $H(w, v)$ is a permutation graph in its two-order, or inversion-graph, presentation [5].

Lemma 2.1 (projection–inversion bridge). *For distinct $x, y \in V$, the equality $\pi_{x,y}(w) = \pi_{x,y}(v)$ holds if and only if no edge of $H(w, v)$ has one endpoint in $\{x\} \times [k]$ and the other in $\{y\} \times [k]$.*

Proof. Within both token orders the occurrences of a fixed letter have the canonical order $(x, 1) < \dots < (x, k)$. The projection onto $\{x, y\}$ is therefore equivalent to the induced total order on the $2k$ tokens in the two fibres: erasing occurrence indices loses no information because equal-letter tokens already have the same order. The two projections are equal precisely when every comparison between an x -token and a y -token agrees in the two orders. This is precisely the absence of a cross-fibre inversion. $\square$

A *compaction* $\phi : H \rightarrow F$ between finite simple graphs is a graph homomorphism that is surjective on vertices and on nonloop edges: for every $xy \in E(F)$, some $uv \in E(H)$ satisfies $\phi(u) = x$ and $\phi(v) = y$ [3].

Theorem 2.2 (characterization). *For a finite simple graph G and an integer $k \geq 1$, the following are equivalent.*

- (i) $G \in \mathcal{G}_k$.
- (ii) There is a permutation graph H and a compaction $\phi : H \rightarrow \overline{G}$ such that $|\phi^{-1}(x)| = k$ for every $x \in V(G)$.
- (iii) There is a poset P of dimension at most two whose ground set is partitioned into k -element chains C_x , indexed by $x \in V(G)$, such that for all distinct $x, y \in V(G)$,

$$xy \in E(G) \iff C_x \cup C_y \text{ is a chain of } P.$$

Proof. Suppose first that $G = G(w, v)$ for two k -uniform words, and take $H = H(w, v)$. Map (x, i) to x . Tokens in one fibre are ordered by their occurrence indices in both words,

so no edge of H is collapsed to a loop. By Lemma 2.1, there is an edge between the x - and y -fibres exactly when $xy \notin E(G)$. Hence the map is a homomorphism to $\overline{G}$, is vertex-surjective, and is edge-surjective; every fibre has size k . This proves (i) $\Rightarrow$ (ii).

Conversely, assume (ii), and realize H by two total orders L_1, L_2 . Every fibre $\phi^{-1}(x)$ is independent in H , since an edge cannot map to a loop in $\overline{G}$. Its tokens consequently have the same relative order in L_1 and L_2 ; label them $(x, 1), \dots, (x, k)$ in that common order. Read L_1 and L_2 from left to right and erase the second token coordinate, producing k -uniform words w, v . The i -th occurrence of x in both words is now the same vertex (x, i) of H , so under this identification $H = H(w, v)$. If $xy \in E(G)$, the homomorphism property forbids every inversion between the two fibres; if $xy \notin E(G)$, edge surjectivity supplies such an inversion. Lemma 2.1 gives, respectively, equal and unequal projections. Thus $G = G(w, v)$, proving (ii) $\Rightarrow$ (i).

For (ii) $\Rightarrow$ (iii), let $P = L_1 \cap L_2$: two tokens are comparable in P when they occur in the same order in both total orders. Its incomparability graph is H , and P has dimension at most two. Each independent fibre of H is a chain of P . For distinct x, y , their two fibres have no incomparability between them exactly when their union is a chain; by the compaction property, this is equivalent to $xy \in E(G)$.

Finally assume (iii). Choose two linear extensions whose intersection is P , let H be the incomparability graph of P , and collapse each C_x to x . The graph H is a permutation graph. Each C_x is a chain and hence independent in H , so no edge becomes a loop. If an H -edge joins C_x to C_y , their union is not a chain, whence $xy \notin E(G)$; the collapse is therefore a homomorphism to $\overline{G}$. Conversely, if $xy \in E(\overline{G})$, then $C_x \cup C_y$ is not a chain, so an incomparable cross-pair supplies a preimage edge. The collapse is vertex- and edge-surjective and has fibres of size k . This proves (iii) $\Rightarrow$ (ii). $\square$

Since $\bigcup_{k \geq 1} \mathcal{G}_k$ contains every finite graph [1, Remark 17], the next minima are finite.

Corollary 2.3. *For every finite graph G , the least k for which $G \in \mathcal{G}_k$ equals the least common fibre size of a uniform compaction from a permutation graph onto $\overline{G}$.*

The characterization also gives a compact worked certificate. The two 2-uniform words

$$w = \text{aabcdbcdee}, \quad v = \text{abcadbcede}$$

represent the graph with edge set $\{ad, ae, bc, be, cd\}$, a 5-cycle. Since $C_5 \notin \mathcal{G}_1$, this exhibits $C_5 \in \mathcal{G}_2 \setminus \mathcal{G}_1$ and may equally be read as a fibre-two compaction certificate for its complement.

3 Neighborhood traces from the two orders

For $A \subseteq V(G)$ and $x \notin A$, the set $N_G(x) \cap A$ is the *neighborhood trace* of x on A .

Theorem 3.1 (trace bound). *Let $G \in \mathcal{G}_k$, let $A \subseteq V(G)$, and put $m = |A|$. Then*

$$|\{N_G(x) \cap A : x \in V(G) \setminus A\}| \leq Q_k(m)^2, \quad Q_k(m) = \binom{k(m+1)}{k}.$$

Proof. Use Theorem 2.2 to choose a permutation graph H , two realizing orders L_1, L_2 , and a fibre- k compaction $\phi : H \rightarrow \overline{G}$. In either order the fibres over A contribute km tokens and therefore cut the order into $km + 1$ gaps. For $x \notin A$, let $d_r(x)$ record, with multiplicity, the gaps of the restriction of L_r to $\phi^{-1}(A)$ occupied by the k tokens of $\phi^{-1}(x)$,

for $r \in \{1, 2\}$. Equivalently, $d_r(x)$ is a weak composition of k into $km + 1$ parts. Hence there are

$$\binom{(km+1)+k-1}{k} = \binom{k(m+1)}{k} = Q_k(m)$$

possible values of each d_r .

The restrictions of L_1, L_2 to the A -fibres are fixed, independently of x . Label the tokens of $\phi^{-1}(x)$ as $x_1 < \dots < x_k$ in their common internal order (the fibre is independent in H). In each L_r , the token x_i occupies the i -th smallest gap recorded by $d_r(x)$, with repetitions counted. Hence the pair $(d_1(x), d_2(x))$ determines every comparison between an x -token and an a -token for each $a \in A$, and therefore determines whether a cross-fibre inversion exists. By the compaction characterization, $ax \in E(G)$ exactly when no such inversion exists. Thus the explicit map

$$x \longmapsto (d_1(x), d_2(x))$$

has a codomain of size $Q_k(m)^2$, and the trace $N_G(x) \cap A$ is a function of its value. The claimed bound follows. $\square$

For fixed k , the bound is $O_k(m^{2k})$. Its point here is not sharpness but the contrast with the 2^m possible subsets of A .

4 An explicit graph outside $\mathcal{G}_2$

At $k = 2$ and $m = 20$,

$$Q_2(20) = \binom{42}{2} = 861, \quad Q_2(20)^2 = 741321.$$

Define a bipartite graph B_{20} with parts

$$A = \{a_0, \dots, a_{19}\}, \quad R = \{b_0, \dots, b_{741321}\}.$$

For $0 \leq i < 20$ and $0 \leq j \leq 741321$, set

$$a_i b_j \in E(B_{20}) \iff \left\lfloor \frac{j}{2^i} \right\rfloor \equiv 1 \pmod{2}.$$

Thus a_0 reads the least significant bit; leading zeros pad every j to 20 bits. The definition is legitimate because $741321 < 2^{20}$.

Theorem 4.1 (explicit obstruction). *The 741342-vertex graph B_{20} is not in $\mathcal{G}_2$.*

Proof. The 741322 integers indexing R have distinct 20-bit expansions, so the traces $N_{B_{20}}(b_j) \cap A$ are pairwise distinct. If $B_{20} \in \mathcal{G}_2$, Theorem 3.1 would permit at most $\binom{42}{2}^2 = 741321$ traces on the fixed 20-set A , a contradiction. $\square$

The hereditariness of $\mathcal{G}_2$ [1, Remark 15] immediately strengthens Theorem 4.1: every graph containing B_{20} as an induced subgraph also lies outside $\mathcal{G}_2$. Moreover, the finite graph B_{20} contains an induced-subgraph-minimal nonmember of $\mathcal{G}_2$, though the proof gives no useful bound on the order of that minimal obstruction. The same trace argument excludes the complement $\overline{B_{20}}$: on A , its traces from the right part are the complements in A of the traces for B_{20} , and hence remain pairwise distinct.

The construction works at every fixed level.

Corollary 4.2 (explicit nonmembers at every level). *Fix $k \geq 1$. For every m satisfying*

$$2^m > \binom{k(m+1)}{k}^2,$$

the bipartite graph with left part $[m]$ and right part any $\binom{k(m+1)}{k}^2 + 1$ distinct subsets of $[m]$, joined by incidence, does not belong to $\mathcal{G}_k$. Such an m exists for every fixed k .

Proof. The selected right vertices have distinct traces on the left part, one more than Theorem 3.1 permits. For fixed k , the square of the binomial coefficient is a polynomial in m of degree $2k$, whereas 2^m eventually exceeds every fixed-degree polynomial. Hence an admissible m exists. $\square$

For scale, Table 1 gives the least m satisfying this particular sufficient inequality for the first five levels. It concerns the coarse bound, not the smallest possible obstruction.

Table 1: Thresholds supplied by the coarse trace bound.

k	least m	right-side vertices	total vertices
1	6	50	56
2	20	741 322	741 342
3	36	49 201 894 226	49 201 894 262
4	53	7 779 446 389 368 901	7 779 446 389 368 954
5	72	2 758 486 094 756 726 663 330	2 758 486 094 756 726 663 402

5 Quantitative calibration

The stars-and-bars bound deliberately forgets correlations among the gap data. To isolate the loss, define $T_k(m)$ to be the maximum number of distinct traces on an m -letter set A obtainable from one fixed pair of k -uniform base words on A by inserting the k occurrences of an outside letter into gaps of each word. Several outside letters with prescribed gap data can be inserted simultaneously: their order inside a common gap does not affect any projection with a letter of A . Conversely, restricting any k -uniform representation to A gives exactly such base words. Therefore $T_k(m)$ is the exact maximum number of neighborhood traces on an m -set in $\mathcal{G}_k$, and

$$T_k(m) \leq \min \{2^m, Q_k(m)^2\}.$$

The least trace-incidence obstruction at a fixed m , if $T_k(m) < 2^m$, has $m + T_k(m) + 1$ vertices.

For $k = 2$, the supplied exact verifier checks directly from the projection definition that the base words

$$\text{eacdfbhidcgaegjfhbij,} \quad \text{afcedbidghjecfabhihg}$$

realize all $2^{10} = 1024$ traces. Thus $T_2(10) = 1024$, and restriction to subsets gives $T_2(m) = 2^m$ for every $m \leq 10$. No obstruction of this trace-incidence form can use $m \leq 10$; Theorem 4.1 shows that one exists at $m = 20$. Consequently the least usable m is at least 11 and at most 20. Determining the intermediate values $T_2(m)$ for $11 \leq m \leq 19$ is left open.

Exploratory local search at $m = 11$ did not find full shattering. As a negative heuristic outcome, this gives no upper bound on $T_2(11)$.

We use the standard open-neighborhood convention for graph VC-dimension [4]: a set $A \subseteq V(G)$ is *shattered* if $\{N_G(x) \cap A : x \in V(G)\} = 2^A$. Let $\text{VCdim}(G)$ be the largest size of such a set and let $\text{VCdim}(\mathcal{G}_2) = \sup_{G \in \mathcal{G}_2} \text{VCdim}(G)$.

Corollary 5.1 (neighborhood VC dimension).

$$10 \leq \text{VCdim}(\mathcal{G}_2) \leq 19.$$

Proof. The certified equality $T_2(10) = 2^{10}$, together with the simultaneous-insertion observation above, gives a member of $\mathcal{G}_2$ whose outside vertices realize every trace on a 10-set. This proves the lower bound.

For the upper bound, shattering is preserved under passing to subsets: if A is shattered and $A' \subseteq A$, then for each $S \subseteq A'$ a vertex satisfying $N_G(x) \cap A = S$ also satisfies $N_G(x) \cap A' = S$. It therefore suffices to rule out a shattered 20-set. For such a set A , Theorem 3.1 bounds the traces contributed by vertices outside A by $Q_2(20)^2 = 741321$, while the vertices of A contribute at most 20 further traces. Thus at most $741321 + 20 = 741341 < 2^{20}$ traces occur, so A is not shattered. $\square$

Three further comparisons prevent the size 741342 from being mistaken for a near-minimality claim.

1. At $k = 1$, C_5 is already a five-vertex nonmember. The coarse trace recipe first applies at $m = 6$ and gives 56 vertices. Exhaustive exact enumeration gives $T_1(4) = 15$, improving this particular trace recipe to 20 vertices. No all- m formula is asserted here.
2. A pair of 2-uniform words on a fixed n -letter alphabet has at most $((2n)!/2^n)^2$ possibilities. Since

$$2^{\binom{n}{2}} > \left(\frac{(2n)!}{2^n} \right)^2$$

first holds at $n = 35$, counting alone proves the existence of a 35-vertex graph outside $\mathcal{G}_2$. This argument supplies no graph.

3. The graph B_{20} is much larger but is canonical at the bit level: its adjacency predicate is a single displayed formula, and its proof requires only the trace bound and exact integer inequalities.

Table 2 summarizes these comparisons by hierarchy level.

Table 2: Obstruction scales, grouped by hierarchy level.			
level	method	vertices	status
$\mathcal{G}_1$	C_5	5	explicit
$\mathcal{G}_1$	exact $T_1(4)$ trace method	20	exhaustive computation
$\mathcal{G}_1$	coarse $Q_1(6)^2$ trace method	56	explicit
$\mathcal{G}_2$	pair counting	35	nonconstructive
$\mathcal{G}_2$	B_{20} from $Q_2(20)^2$	741342	explicit

6 Verification and formal scope

The public companion artifact has four independent, standard-library Python entry points. The first recomputes the binomial coefficients, threshold minimality, level table, 35-vertex crossover, and VC-dimension endpoint using exact integers and a Pascal-triangle implementation independent of `math.comb`. The second forms projections literally to check the C_5 and $T_2(10)$ certificates and exactly enumerates $T_1(4)$. The third constructs one simultaneous 2-uniform pair containing all 1024 outside letters and verifies their traces literally. The fourth compares the word and compaction descriptions on small instances; its genuine test is equality of the resulting isomorphism classes, with separate guards for the homomorphism and edge-surjectivity axioms. Failures use explicit exceptions rather than `assert`, so optimization with `python -O` cannot disable them.

The Lean 4 companion is intentionally scoped. It defines the inversion graph of two genuine strict linear orders, proves for distinct fibres that agreement of every cross-token comparison is equivalent to the absence of an inversion edge, and checks the exact arithmetic (including the VC-dimension endpoint), the injection into 20-bit codes, and the finite pigeonhole endpoint. It uses neither `native_decide` nor any project axiom. It does *not* formalize word lists, the full compaction equivalence, stars and bars, or Theorem 4.1; those remain the ordinary proofs above. The archive pins Lean 4 and mathlib, records the build commands and axiom audit, and includes the resulting logs.

7 Discussion

Theorem 2.2 turns equality of word projections into a standard two-order object with prescribed fibres. It may be useful for recognition algorithms, smaller obstruction searches, and the conjectured properness of every inclusion $\mathcal{G}_k \subseteq \mathcal{G}_{k+1}$, where the inclusion itself is due to Adamson et al. [1, Remark 16 and Conj. 31].

There is also a limited geometric reading. In a segment model of the permutation graph, a fibre for $k = 2$ consists of two noncrossing segments; their convex hull is a trapezoid. A crossing between lateral sides makes the two trapezoids intersect, so the compaction target is a spanning subgraph of that trapezoid intersection graph. The converse fails: trapezoids may also intersect by containment or partial overlap. For example, rectangles with intervals $[0, 2]$ and $[1, 3]$ on both parallel lines overlap although no lateral sides cross and neither rectangle contains the other. Thus one cannot describe the target merely by deleting containment edges.

Hakim and Bayzid use the same broad neighborhood-capacity/incidence-obstruction strategy for threshold pairwise-compatibility graphs [6, Sec. 6]. Their tree-metric class and $(Cm^3)^k$ bound differ from the two-word gap code and $Q_k(m)^2$ bound here.

The largest quantitative gap is between the 35-vertex existence proof and the 741342-vertex explicit graph. Concrete next problems are to find a small graph outside $\mathcal{G}_2$, determine $\text{VCdim}(\mathcal{G}_2)$ exactly and the intermediate values $T_2(m)$ for $11 \leq m \leq 19$, recognize fixed-fibre compaction images efficiently, and decide whether every inclusion $\mathcal{G}_k \subseteq \mathcal{G}_{k+1}$ is proper.

Two concrete questions.

1. Does B_{20} belong to $\mathcal{G}_3$? The trace bound does not exclude this: $Q_3(20) = 39711$, so $Q_3(20)^2 > 741322$. An affirmative answer would explicitly witness $\mathcal{G}_2 \subsetneq \mathcal{G}_3$, the $k = 2$ case of the hierarchy conjecture in [1, Conj. 31].

2. Is $\mathcal{G}_2$ closed under graph complementation? The class $\mathcal{G}_1$ is complement-closed by reversing one representing permutation [1, Remark 9], but for higher uniformity reversal is no longer a binary choice between the only two projected orders.

Data and code availability

The LaTeX source, exact Python verifiers, scoped Lean 4 source, build logs, and checksums are archived at Zenodo.

DOI: [10.5281/zenodo.21986230](https://doi.org/10.5281/zenodo.21986230).

AI-use disclosure

OpenAI Codex and Anthropic Claude were used extensively throughout this project. Their use was substantive and pervasive, not limited to copyediting: it included the literature and scoop sweeps, proof development and checking, a dedicated adversarial review of an earlier draft, analysis and resolution of that review, Lean design and debugging, Python verifier development, LaTeX drafting and revision, citation and metadata work, and release packaging and quality assurance. The author made the final mathematical and editorial decisions and assumes responsibility for the claims, citations, artifacts, and any remaining errors.

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